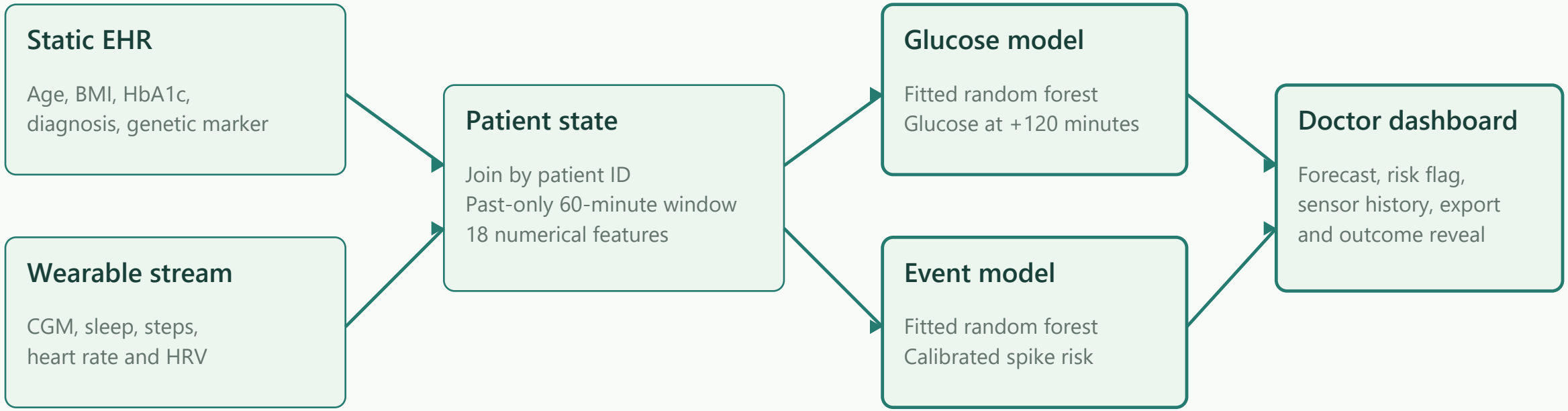
Digital Twin system architecture

Runtime path: two data streams to a patient-specific state and two-hour prediction



Future observations are held back from inference. They are revealed only to check the saved prediction.

Offline training and evaluation

Cohort	Patients	Use
Training	54	Fit regression and classifier candidates
Validation	12	Select models and alert cutoff
Calibration	12	Fit risk mapping and residual band
Final test	12	Score 4,416 rolling forecasts

Python exports fitted trees and calibration parameters to JSON. JavaScript performs inference in the browser. Separate CSV streams and every test prediction are included in the repository.